Social Media Marketing Campaign Analytics

Executive Report - From Raw Data to Budget Decisions

Analysis of 6,000+ social media campaigns answering one business question: how can a company improve campaign performance and use its marketing budget more efficiently?

Tools: Python (Pandas, NumPy, Matplotlib), SQL (SQLite), Looker Studio.

1. Executive Summary

This report analyzes 6,000 cleaned social media campaigns (from 6,200 raw records) to find where marketing budget works hardest. Three headline results:

1. Email is the most efficient platform (ROAS 4.17); low-spend campaigns dominate efficiency. **2.** The best content combination is Email + Video (ROAS 4.66); the most-seen content (Video, 395M impressions) is not the most profitable. **3.** Audience segments show no meaningful gaps, so budget should not move by age, location or device.

Star finding: campaigns spending under $1k concentrate just 3.9% of the budget and generate 19.0% of total revenue.

Business implication: efficiency lives at low spend. A modeled scenario moving 20% of the highest-spend range budget to low-spend efficiency lifts total revenue from $57.15M to $73.58M (+28.8%) under explicit constant-efficiency assumptions.

2. Business Objective

General question: how can a company improve social media campaign performance and use its marketing budget more efficiently? It was broken into five questions: (1) platform efficiency, (2) content performance, (3) budget efficiency and diminishing returns, (4) audience segmentation, (5) budget reallocation opportunity.

3. Dataset & Methodology

Source: raw Excel file with 6,200 records and 16 columns (campaign, platform, dates, spend, impressions, clicks, precomputed CTR/CPC, conversions, revenue, audience, device, post type). Workflow: data quality audit (no changes), data cleaning on a working copy, KPI rebuild from definitions, exploratory analysis, five business questions, then SQL staging (SQLite) reproducing the key cuts as an independent check. All ratios are computed on totals (overall), never averaged.

4. Data Quality & Cleaning

Audit found: 200 exact duplicate rows (which exactly matched 200 repeated Campaign_IDs, verified NaN-aware) - removed, leaving 6,000 rows with unique IDs. Platform had 14 label variants of 6 real platforms (case/space variants plus 'IG') - normalized with an explicit map. Precomputed CTR/CPC columns matched recalculation in only 3 of 5,422 and 6 of 5,422 rows - dropped as unreliable. Dates were already datetime, but 2,739 rows have End before Start: kept and documented, since deleting 46% was unjustifiable. Missing values (~300 each in Spend, Clicks, Conversions, Revenue, randomly spread) were kept as NaN: imputing money metrics would invent results. No negatives, no IQR outliers, 271 rows with Clicks above Impressions kept and flagged. Clicks/Conversions use nullable integers.

5. Key Performance Indicators

Rebuilt from definitions (ratios stored as decimals): CTR = Clicks/Impressions, CPC = Spend/Clicks, Conversion Rate = Conversions/Clicks, CPA = Spend/Conversions, ROAS = Revenue/Spend, ROI = (Revenue-Spend)/Spend. Overall totals: Spend $14.36M, Revenue $57.15M, 1.51B impressions, 141.4M clicks, 14.2M conversions, overall ROAS 3.98, overall CPA $1.01. Note: mean ROAS (9.48) is inflated by outliers, which is why every comparison uses overall ratios. One CPA case (zero conversions) is NaN.

Metric	Definition
CTR	Clicks / Impressions
CPC	Spend / Clicks
Conversion Rate	Conversions / Clicks
CPA	Spend / Conversions
ROAS	Revenue / Spend
ROI	(Revenue - Spend) / Spend

6. Findings

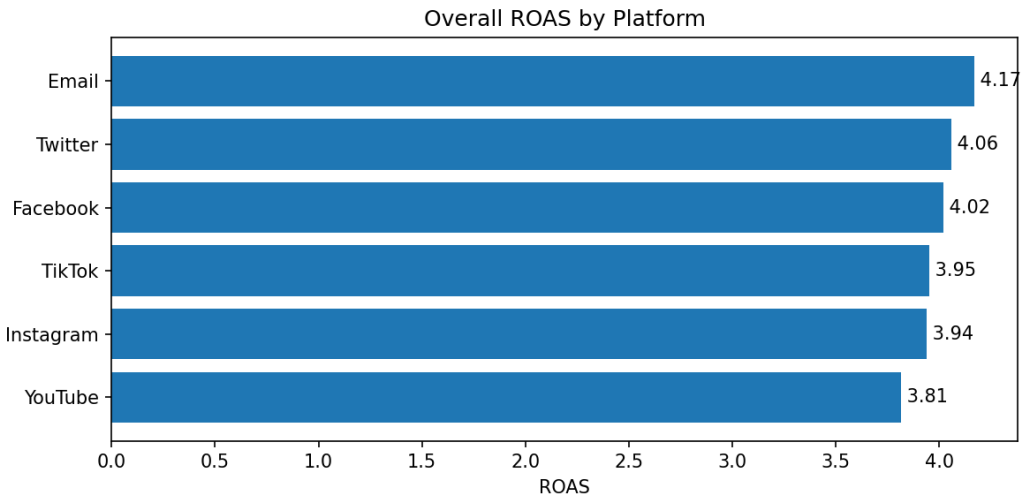
1. Email leads efficiency (ROAS 4.17, ROI 3.17); YouTube trails (3.81). 2. Instagram runs 2x the campaigns (1,739) but mid-table ROAS (3.94): volume is not efficiency. 3. Text is the best content (4.08); Video leads impressions (395M) but not return (4.01). 4. Email + Video is the best combination (4.66); YouTube + Image the worst (3.40). 5. Carousel converts best (10.51%) but monetizes average (ROAS 3.95). 6. The 0-1k spend range holds 3.9% of spend and 19.0% of revenue (ROAS 18.39) vs 2.11 for 4k-5k. 7. Segments are flat (age 3.83-4.10, location 3.90-4.04, device 3.95-4.02). 8. Platforms are budget-balanced (gaps under 0.006). 9. Modeled reallocation scenario: +28.8% revenue under stated assumptions.

7. Answers to the 5 Business Questions

Q1 - Platform efficiency

Email delivers the best balance (ROAS 4.17, ROI 3.17, CPA 0.96), Twitter follows (ROAS 4.06, lowest CPA 0.95). Instagram dominates volume without leading efficiency.

Platform	Campaigns	Spend $M	Revenue $M	ROAS	CPA	ROI
Email	831	1.96	8.18	4.17	0.96	3.17
Twitter	812	1.88	7.61	4.06	0.95	3.06
Facebook	854	2.03	8.17	4.02	1.01	3.02
TikTok	855	2.08	8.21	3.95	1.02	2.95
Instagram	1,739	4.20	16.53	3.94	1.06	2.94
YouTube	909	2.21	8.44	3.81	1.03	2.81



Q2 - Content performance

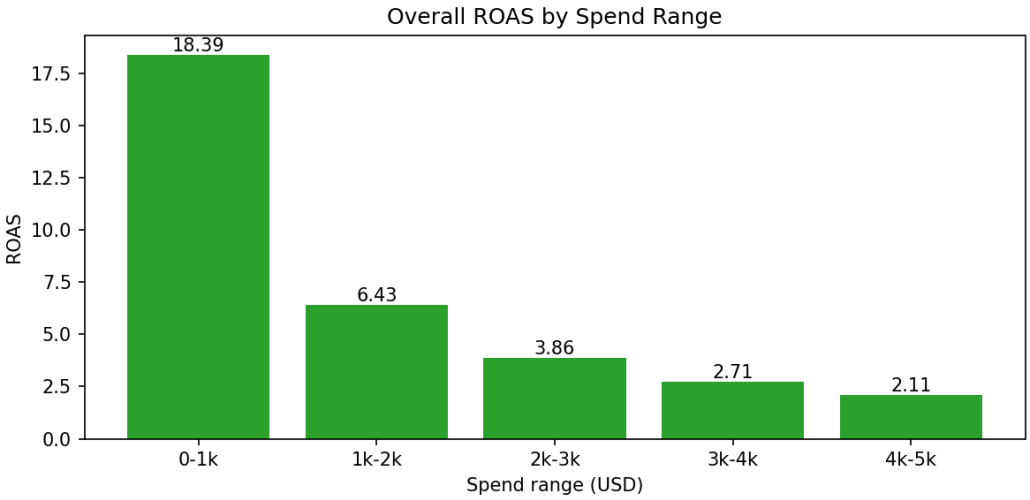
Text wins overall (4.08); the most-impressed content does not win. Email + Video is the top combination (4.66); YouTube + Image needs fixing before funding (3.40).

Post_Type	Campaigns	Impressions M	ROAS	CTR	Conv. Rate
Text	1,498	375.0	4.08	9.38%	10.16%
Video	1,572	395.7	4.01	9.59%	9.79%
Carousel	1,504	382.3	3.95	9.03%	10.51%
Image	1,426	355.3	3.86	9.51%	9.73%

Q3 - Budget efficiency

Efficiency collapses as spend grows: ROAS 18.39 below $1k down to 2.11 above $4k, while revenue per range stays near-flat (~$10-11M). The honest read: returns are flat, not classically diminishing - a fingerprint of synthetic data, reported as observed with this caveat.

Range	Campaigns	Spend share	Revenue share	ROAS	CPA
0-1k	1,088	3.9%	19.0%	18.39	0.22
1k-2k	1,140	11.9%	20.2%	6.43	0.62
2k-3k	1,184	20.6%	21.0%	3.86	1.06
3k-4k	1,164	28.4%	20.3%	2.71	1.47
4k-5k	1,124	35.1%	19.5%	2.11	1.93

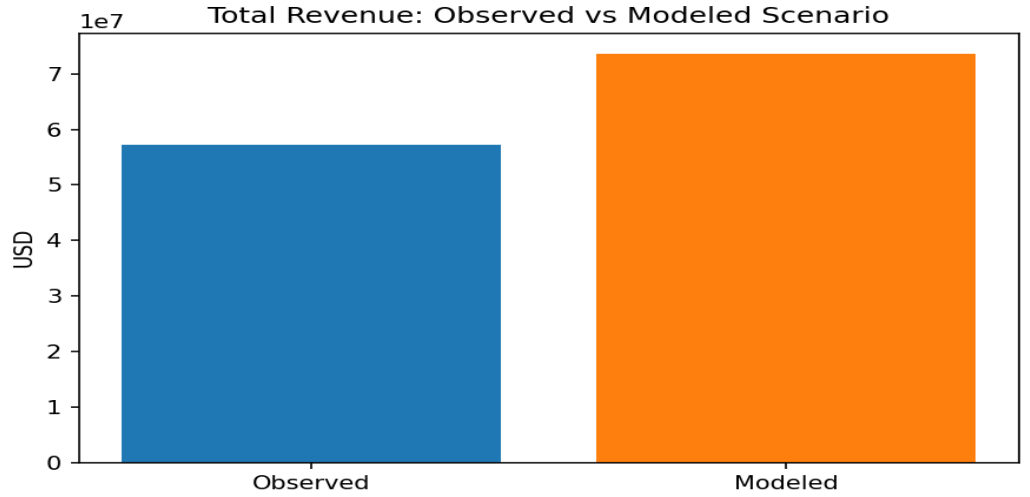


Q4 - Audience and segmentation

No actionable gaps: best age 35-44 (4.10) vs worst 25-34 (3.83); best location Toronto (4.04, also best conversion rate 10.76%) vs London (3.90); best device Tablet (4.02) vs Desktop (3.95). Recommendation: do not move budget by segment.

Q5 - Reallocation opportunity

Observed: platform revenue share tracks spend share within 0.006 - misallocation lives in spend size, not channel. Modeled scenario (assumptions: constant bin ROAS, no saturation): moving 20% of the 4k-5k spend ($1.01M) to low-spend efficiency lifts revenue from $57.15M to $73.58M (+28.8%). A scenario, not a promise.



8. Key Opportunity

Campaigns spending under $1k concentrate just 3.9% of the budget and generate 19.0% of total revenue.

Confirmed on fully complete rows (ROAS 19.44 vs 18.39). The opportunity is structural: pilot low-spend campaign patterns with controlled budget shifted from the 4k-5k range, measuring real ROAS before

scaling.

9. Recommendations

1. Cap high-spend campaigns, test low-spend patterns (Q3: ROAS 18.39 vs 2.11). **2. Prioritize Email + Video and Facebook + Text for conversion goals** (ROAS 4.66, 4.64). **3. Fix YouTube + Image before funding it** (ROAS 3.40). **4. Do not reallocate by segment** (noise-level gaps). **5. Report overall ratios, never averages** (mean ROAS 9.48 vs overall 3.98).

10. Limitations

Educational/portfolio dataset: synthetic, metrics independent of spend (correlations near zero), so effect sizes describe this data, not market laws. Dates unreliable as ranges (46% inverted). 271 rows with Clicks above Impressions kept and flagged. Missing values kept as NaN with a complete-case robustness check. The +28.8% is a modeled scenario under stated assumptions. Correlation is not causation.

11. Final Conclusion

The data tells one coherent story: efficiency concentrates in low spend, Email-led mixes and Text/Video content, while volume, segments and extra dollars add little. A marketing team acting on this would cap high-spend campaigns, fund Email + Video, fix YouTube + Image, and measure everything with overall ratios. That is how raw campaign data becomes budget decisions.